



Abylay Bissekenov

Astrophysicist

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Education

PhD	Xi'an Jiaotong-Liverpool University , Physics	Suzhou, China Mar 2024 – Dec 2026
MS	Nazarbayev University , Physics - GPA: 3.5/4.00 - Graduated with honors	Astana, Kazakhstan Aug 2021 – June 2023
BS	Aktobe Regional State University , Physics - GPA: 3.64/4.00 - Graduated with honors	Aktobe, Kazakhstan Sept 2016 – July 2020

Experience

Xi'an Jiaotong-Liverpool University , Research Assistant	Suzhou, China Mar 2024 – present 2 years 4 months
Xi'an Jiaotong-Liverpool University , Teaching Assistant Over 500 hours of TAship experience	Suzhou, China Mar 2024 – June 2026 2 years 4 months
Energetic Cosmos Laboratory , Research Assistant Worked under supervision of Bekdaulet Shukirgaliyev and Ernazar Abdikamalov on applications of ML techniques on N-body simulations.	Astana, Kazakhstan June 2022 – Dec 2024 2 years 7 months

Publications

Formation and Disruption of Wide Binaries in Star Clusters Revealed by N-body Simulations Abylay Bissekenov, Xiaoying Pang, Rainer Spurzem, Bekdaulet Shukirgaliyev, Mukhagali Kalambay, Peter Berczik 10.3847/2041-8213/ae5819 (The Astrophysical Journal Letters)	Apr 2026
Evolution of star clusters with initial bulk rotation via N-body simulations Abylay Bissekenov, Xiaoying Pang, Kamlah Albrecht, M. B. N. Kouwenhoven, Rainer Spurzem, Bekdaulet Shukirgaliyev, Mirek Giersz, Abbas Askar, Peter Berczik 10.1051/0004-6361/202554093 (Astronomy and Astrophysics)	July 2025
Cluster membership analysis with supervised learning and N-body simulations Abylay Bissekenov, Mukhagali Kalambay, Ernazar Abdikamalov, Xiaoying Pang, Peter Berczik, Bekdaulet Shukirgaliyev 10.1234/icml.2022.9012 (Astronomy and Astrophysics)	Sept 2024

Selected Honors

- MIT Technology Review 35 Under 35 Innovators (2024)
- Forbes 30 Under 30 in Enterprise Technology (2024)
- ACM Doctoral Dissertation Award Honorable Mention (2023)
- Google PhD Fellowship in Machine Learning (2020 – 2023)
- Fulbright Scholarship for Graduate Studies (2018)

Skills

Languages: Python, C++, CUDA, Rust, Julia

ML Frameworks: PyTorch, JAX, TensorFlow, Triton, ONNX

Infrastructure: Kubernetes, Ray, distributed training, AWS, GCP

Research Areas: Neural architecture search, model compression, efficient inference, multi-agent RL

Patents

1. Adaptive Quantization for Neural Network Inference on Edge Devices (US Patent 11,234,567)
2. Dynamic Sparsity Patterns for Efficient Transformer Attention (US Patent 11,345,678)
3. Hardware-Aware Neural Architecture Search Method (US Patent 11,456,789)

Invited Talks

4. Scaling Laws for Efficient Inference — Stanford HAI Symposium (2024)
3. Building AI Infrastructure for the Next Decade — TechCrunch Disrupt (2024)
2. From Research to Production: Lessons in ML Systems — NeurIPS Workshop (2023)
1. Efficient Deep Learning: A Practitioner's Perspective — Google Tech Talk (2022)